Assignment 1: Scheduling and Workflow Automation Using Zapier

Introduction

The objective of this assignment is to develop an automated invoice management system using Zapier, Google Sheets, and Gmail. The system is designed to reduce manual effort in sending payment reminders and updating invoice records after receiving payment confirmations.

The workflow consists of two separate Zaps. The first Zap retrieves pending invoices from Google Sheets and sends payment reminder emails. The second Zap monitors incoming email replies, identifies payment confirmations, and updates the corresponding invoice records.

By integrating these applications, the solution demonstrates how workflow automation can improve the efficiency, accuracy, and tracking of routine financial operations.

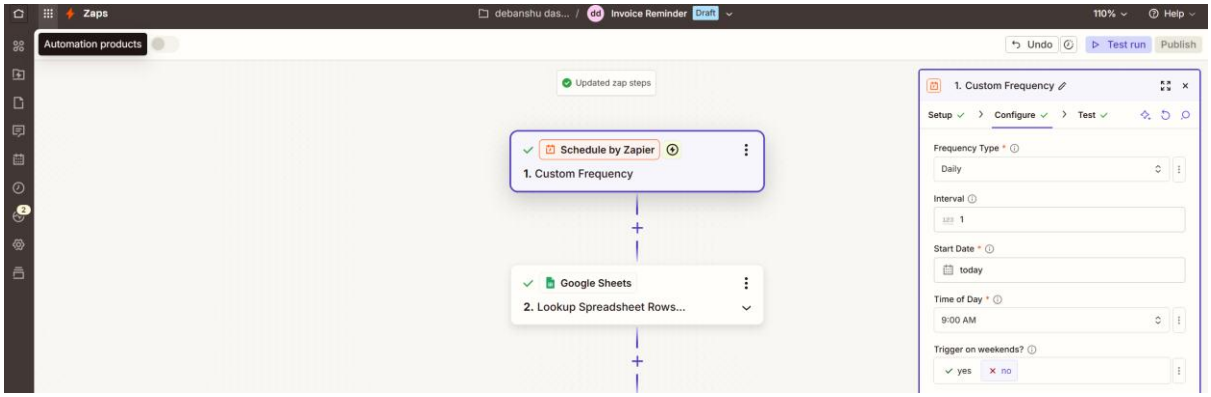
Zap 1: Payment Request Automation

Step 1 — Schedule by Zapier

Schedule by Zapier was configured as the trigger for the payment reminder workflow. The trigger was set to run at a predefined time so that the invoice records could be checked automatically without requiring manual initiation each day.

When the scheduled trigger runs, it starts the workflow and passes control to the subsequent steps. This allows the system to periodically identify invoices that still require payment reminders.

The scheduled trigger establishes the starting point of the automation and supports a consistent reminder process.



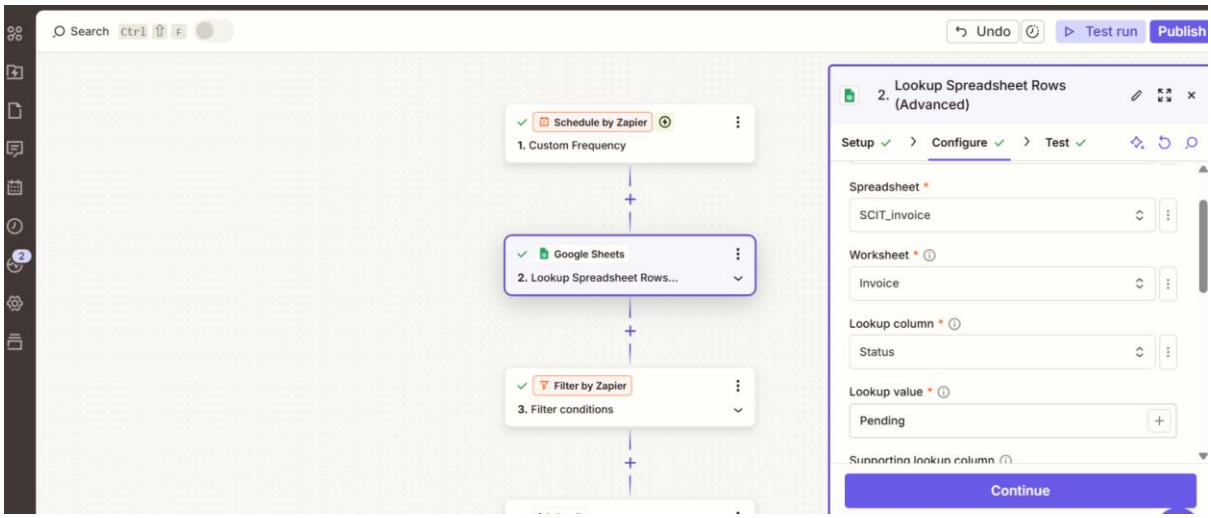
Step 2 — Google Sheets: Lookup Spreadsheet Rows

Google Sheets was connected to the workflow as the primary source of invoice information. The spreadsheet contains invoice details such as Invoice ID, Type, Name, Amount, Account Number, Bank Name, and Status.

The lookup action was configured to identify records whose payment status is marked as **Pending**. This helps distinguish invoices that require follow-up from those that have already been completed.

For the sample dataset, the pending invoices are 1001, 1002, and 1003. Invoices 1004 and 1005 are already marked as completed and should not receive further payment reminders.

The output of this step provides the invoice information required by the subsequent workflow actions.



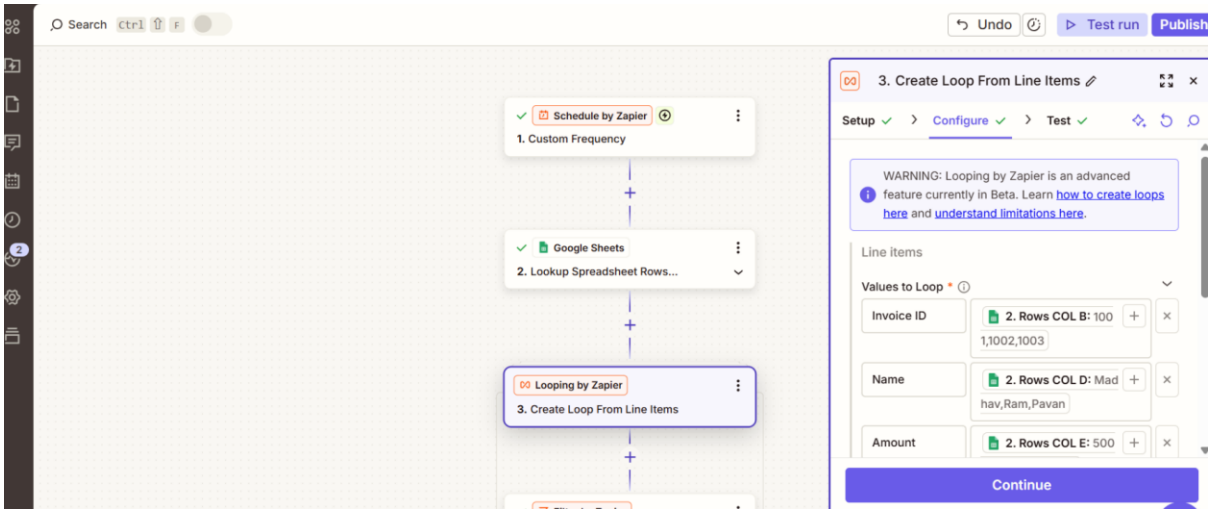
Step 3 — Looping by Zapier: Create Loop From Line Items

Looping by Zapier was used to process the retrieved invoice records individually. The relevant invoice fields, including Invoice ID, Type, Name, Amount, Account Number, Bank Name, and Status, were mapped into the loop configuration.

The purpose of this step is to ensure that each pending invoice is handled as a separate item. The loop maintains the relationship between an invoice and its associated details so that the correct information is inserted into each reminder email.

For example, invoice 1001 contains the details for Madhav, invoice 1002 contains the details for Ram, and invoice 1003 contains the details for Pavan. Each invoice must retain its own amount and banking information throughout the process.

This approach supports the processing of multiple pending invoices within a single scheduled workflow.



Step 4 — Gmail: Send Payment Reminder Email

Gmail was integrated into the workflow to send personalized payment reminder emails using the invoice details retrieved from Google Sheets.

The email subject and body were configured to include dynamic values from the current invoice record. The message contains the Invoice ID, customer name, invoice type, outstanding amount, payment status, bank name, and account number.

The reminder requests that the recipient review and process the pending payment. It also asks the recipient to reply with the Invoice ID and Transaction ID/UTR if payment has already been completed.

The dynamic field mapping allows the same email template to be reused for different invoices without manually entering the details for each message.

The recipient is selected according to the invoice type: Project invoices are routed to the designated Project recipient, while Maintenance invoices are routed to the designated Maintenance recipient.

Sample email format

Subject: Payment Reminder – Invoice [Invoice ID]

Dear [Recipient Name],

This is a friendly reminder that payment for the following invoice is currently pending.

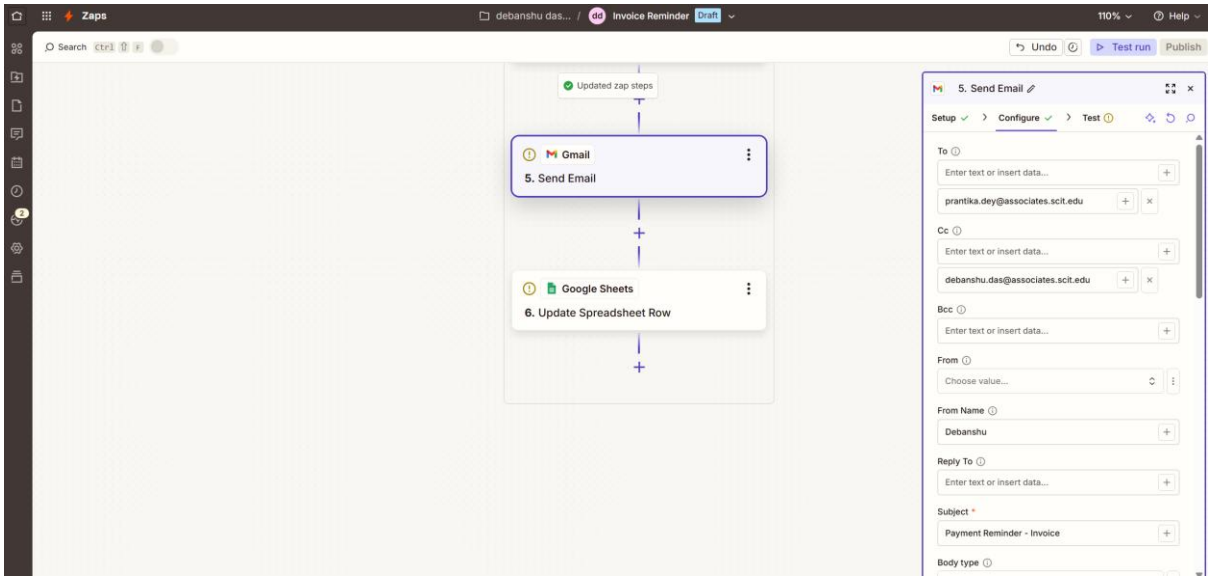
- Invoice ID: [Invoice ID]
- Name: [Name]
- Type: [Type]
- Amount Due: ₹[Amount]
- Payment Status: [Status]

Bank Details:

- Bank Name: [Bank Name]
- Account Number: [Account Number]

Kindly review and process the payment. If the payment has already been completed, please reply to this email with the Invoice ID and Transaction ID/UTR for reconciliation.

Regards,
Accounts Team



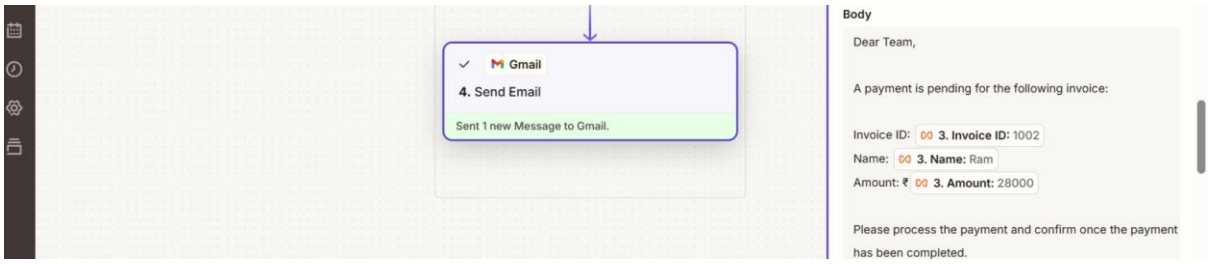
Step 5 — Testing and Validation of Zap 1

The first Zap was tested to verify the connection between the scheduled trigger, Google Sheets lookup, looping action, and Gmail email action.

The test checks whether the workflow identifies the pending invoices, processes the invoice records individually, and inserts the corresponding values into the email template.

The test also helps identify configuration issues, such as incorrect field mapping or missing dynamic values, before the workflow is activated.

The expected result is that only pending invoices are selected and that each reminder contains the correct information for its corresponding invoice.



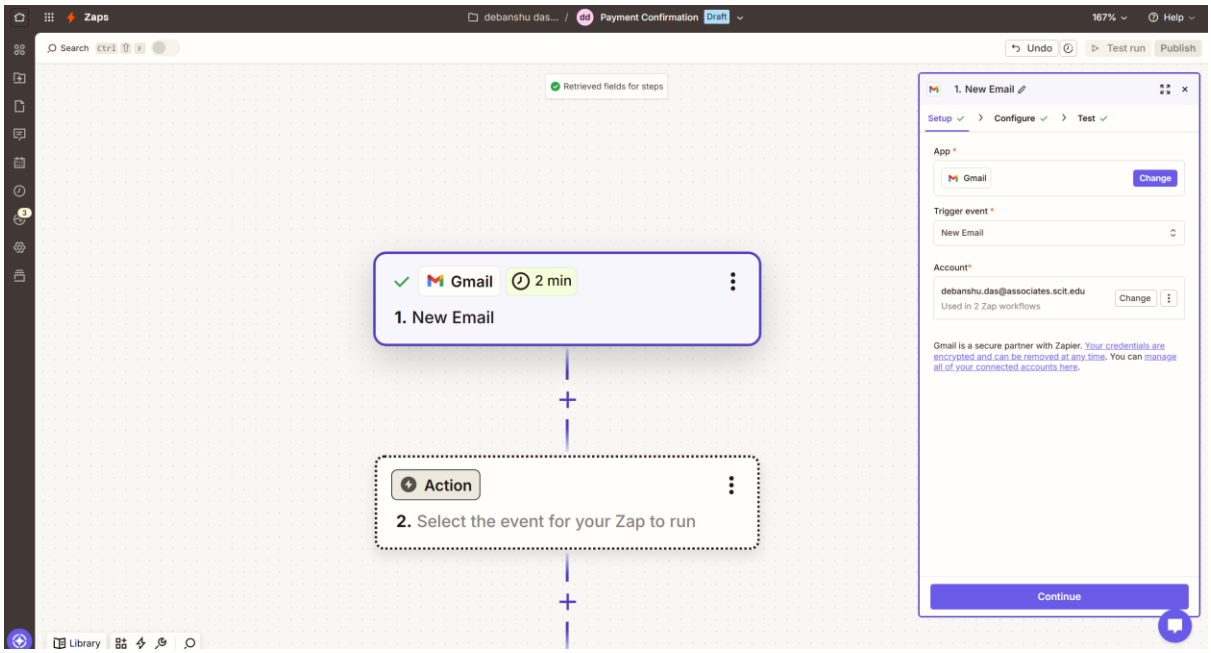
Zap 2: Automated Payment Confirmation and Reconciliation

Step 1 — Gmail: New Email Trigger

A separate Zap was created using Gmail's New Email trigger to detect incoming replies that may contain payment confirmations.

This trigger monitors the configured Gmail mailbox and passes incoming email information to the next workflow step. The email subject and body provide the context required to determine whether the message relates to a previously issued payment reminder.

Using a separate Zap for payment confirmation allows the invoice status update process to operate independently of the scheduled reminder workflow.

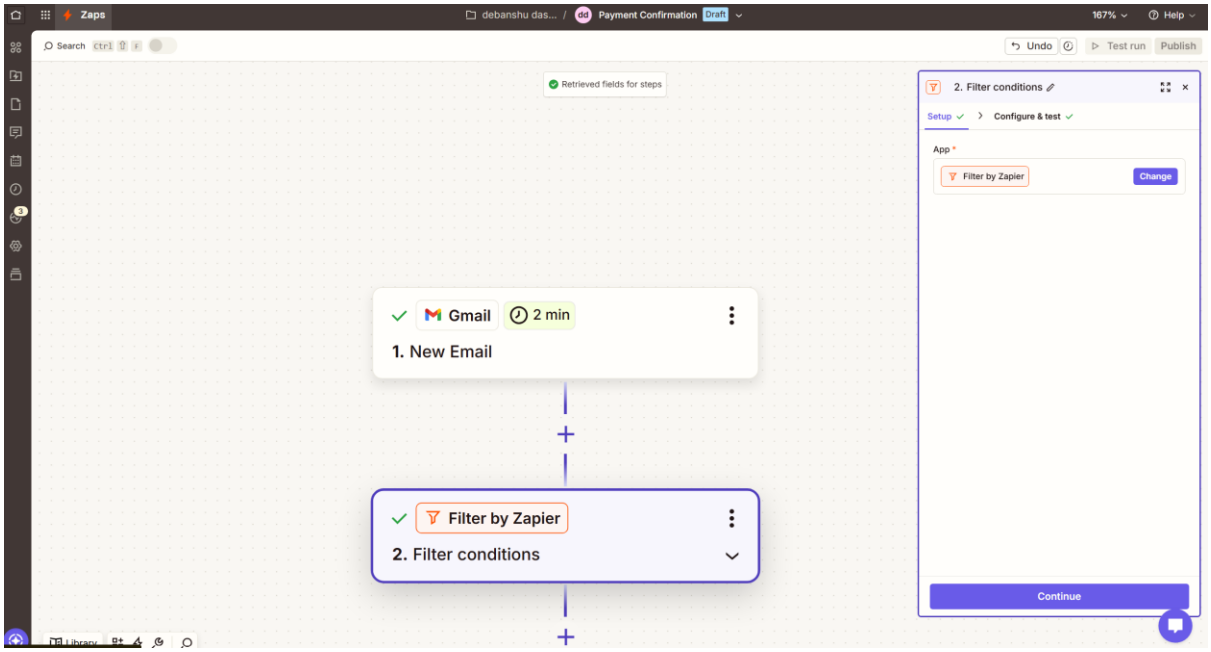


Step 2 — Filter by Zapier: Identify Payment Confirmations

Filter by Zapier was added to prevent unrelated emails from proceeding through the payment confirmation workflow.

The filter checks the incoming email for a payment confirmation phrase, such as “Payment Done.” Only messages satisfying the configured condition are allowed to continue to the next step.

This condition helps reduce the possibility of unrelated emails triggering invoice updates. Messages indicating that payment will be made later or that payment is still being processed should not be treated as completed payments.



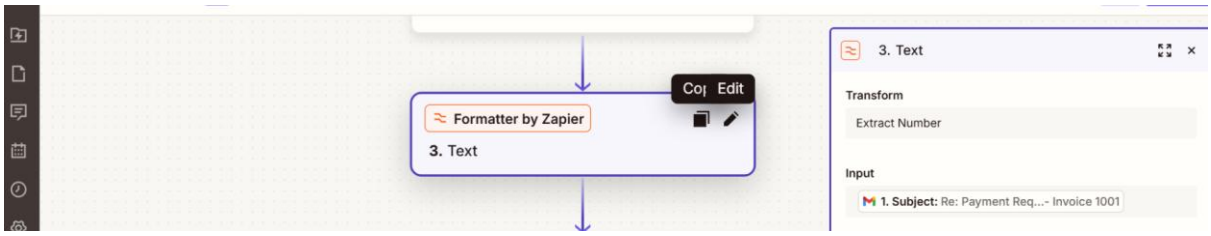
Step 3 — Formatter by Zapier: Extract Invoice ID

Formatter by Zapier was configured to extract the Invoice ID from the subject of the payment confirmation email.

For example, a reply with the subject “Payment Reminder – Invoice 1002” can be processed to identify invoice 1002. The extracted value is then passed to the Google Sheets lookup action.

This step links the incoming confirmation email to the corresponding invoice record without requiring the Invoice ID to be entered manually during reconciliation.

If the Invoice ID cannot be extracted reliably from the subject, the email body or conversation context must be checked before proceeding.



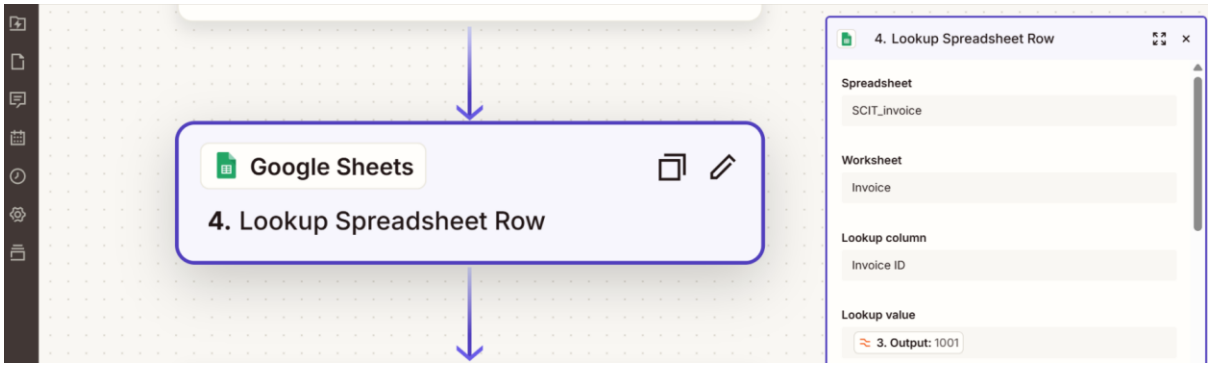
Step 4 — Google Sheets: Lookup Spreadsheet Row

Google Sheets was used to locate the invoice record associated with the extracted Invoice ID.

The lookup action searches the invoice worksheet using the Invoice ID as the matching value. When a matching record is found, the workflow obtains the corresponding spreadsheet row so that the update can be applied to the correct invoice.

This step connects the payment confirmation received by email to the relevant record in the invoice tracking spreadsheet.

If the Invoice ID does not match a record, the workflow should not update an unrelated invoice.



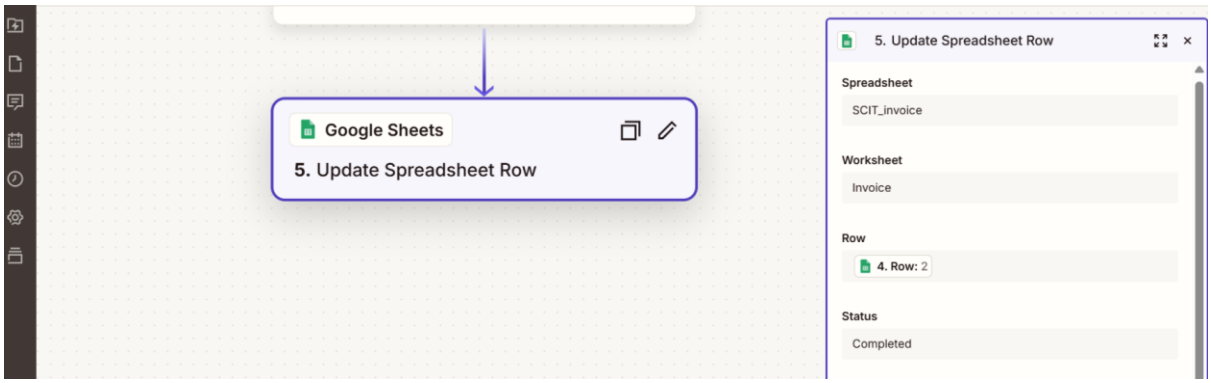
Step 5 — Google Sheets: Update Spreadsheet Row

After identifying the matching invoice, the Google Sheets update action changes its payment status from Pending or Reminder Sent to Completed.

The update is applied to the corresponding row, ensuring that the central invoice tracking sheet reflects the latest payment information.

For the extended version of the workflow, a Transaction ID column is also maintained. When the reply contains a valid Transaction ID/UTR, that value is saved in the corresponding invoice record along with the Completed status.

The transaction reference supports payment reconciliation and provides a reference for subsequent verification. If the confirmation is ambiguous, the Invoice ID cannot be matched, or the Transaction ID is missing, the invoice should remain unchanged until the information is verified.



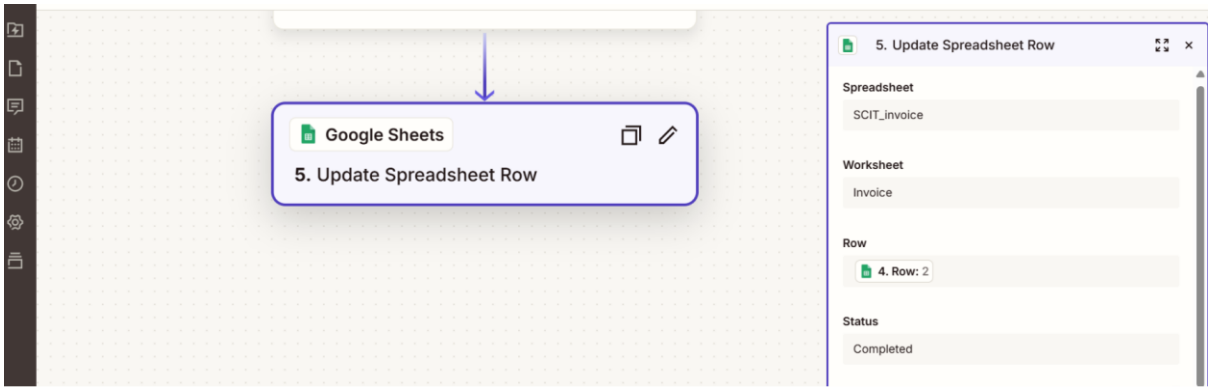
Step 6 — Testing and Validation of Zap 2

The second Zap was tested using a payment confirmation email to verify the flow from Gmail through the filter, Formatter, Google Sheets lookup, and update action.

The test checks whether the confirmation message satisfies the filter condition, whether the Invoice ID is extracted correctly, and whether the corresponding spreadsheet row is identified.

The final validation checks that the status changes to Completed and that the Transaction ID/UTR is recorded in the correct row when provided.

Testing with different email scenarios helps confirm that unrelated messages, incomplete confirmations, and unmatched Invoice IDs do not cause incorrect spreadsheet updates.



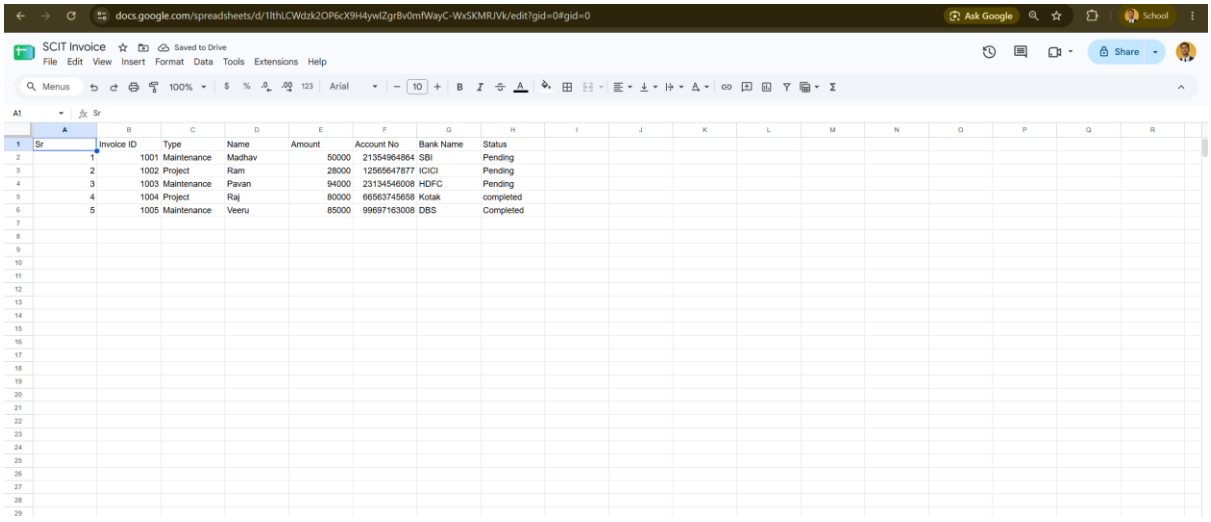
Conclusion

This assignment demonstrates how Zapier can connect scheduling, spreadsheet management, and email communication to automate invoice follow-up and payment reconciliation.

The first Zap automates the identification of pending invoices and the generation of personalized payment reminders. The second Zap processes incoming payment confirmations and updates the corresponding invoice records.

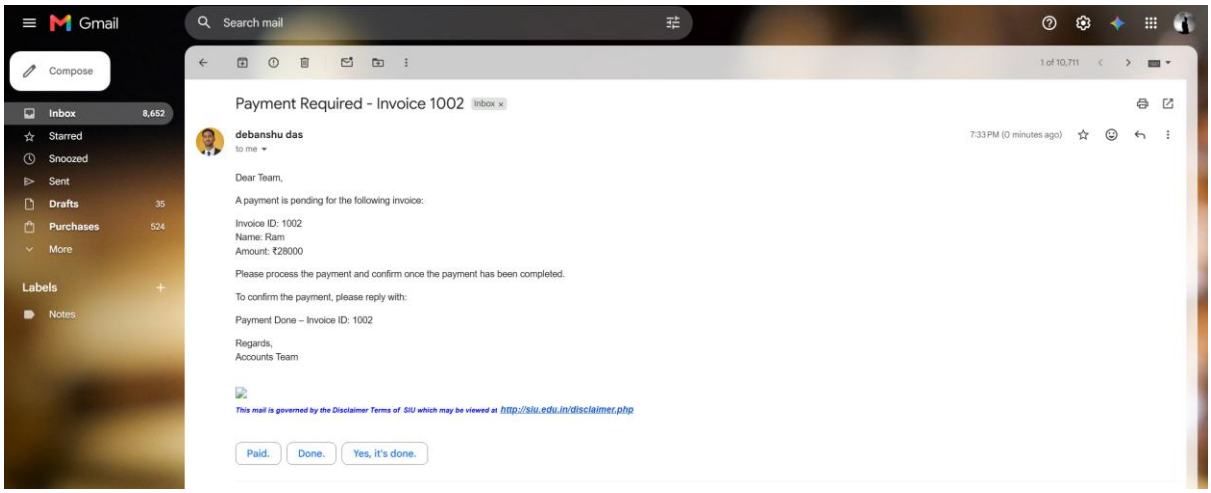
The integration of Google Sheets and Gmail reduces repetitive manual work and maintains a centralized record of invoice statuses. The additional Transaction ID field further supports reconciliation by associating payment confirmations with their corresponding invoice records.

Overall, the solution demonstrates the practical application of workflow automation in improving the consistency, traceability, and efficiency of invoice management.



SCIT Invoice

Sr	Invoice ID	Type	Name	Amount	Account No	Bank Name	Status
1	1001	Maintenance	Madhav	50000	21354964864	SBI	Pending
2	1002	Project	Ram	28000	12565647877	ICICI	Pending
3	1003	Maintenance	Pavan	94000	23134546008	HDFC	Pending
4	1004	Project	Raj	80000	66563745658	Kotak	completed
5	1005	Maintenance	Veeru	85000	99697163008	DBS	Completed



Payment Required - Invoice 1002

debanshu das
to me

7:33 PM (0 minutes ago)

Dear Team,

A payment is pending for the following invoice:

Invoice ID: 1002
Name: Ram
Amount: ₹28000

Please process the payment and confirm once the payment has been completed.

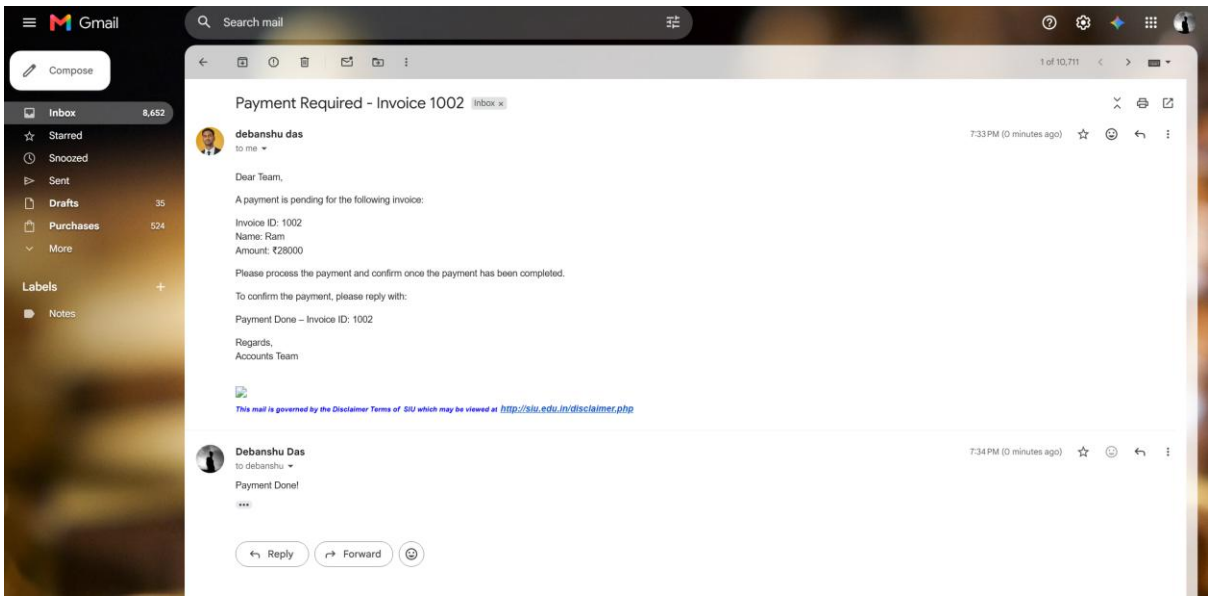
To confirm the payment, please reply with:

Payment Done – Invoice ID: 1002

Regards,
Accounts Team

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Paid. Done. Yes, it's done.



Payment Required - Invoice 1002

debanshu das
to me

7:33 PM (0 minutes ago)

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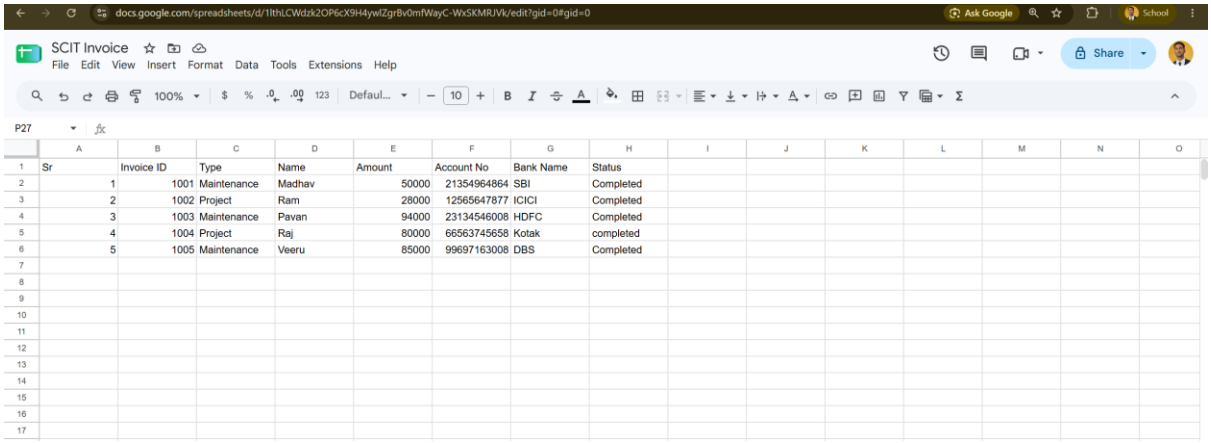
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Debanshu Das
to debanshu

7:34 PM (0 minutes ago)

Payment Done!

Reply Forward



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Sr	Invoice ID	Type	Name	Amount	Account No	Bank Name	Status							
2	1	1001	Maintenance	Madhav	50000	21354964864	SBI	Completed							
3	2	1002	Project	Ram	28000	12565647877	ICICI	Completed							
4	3	1003	Maintenance	Pavan	94000	23134546008	HDFC	Completed							
5	4	1004	Project	Raj	80000	66563745658	Kotak	completed							
6	5	1005	Maintenance	Veeru	85000	99697163008	DBS	Completed							
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